

Airbrakes CFD Report

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Table of Definitions	3
Acknowledgements	4
0.0 Introduction	5
0.1 Problem Description	5
0.2 Background	5
1.0 Scope and Requirements	6
1.1 Scope	6
1.2 Constraints	6
1.2.2 Team Constraints	6
1.2.3 Project Constraints	7
1.3 Interfaces	7
1.4 Timeline	8
2.0 Apparatus	8
2.1 Preparing Geometry	8
2.2 Meshing	9
2.3 Solving	10
2.4 Solution and Post Processing	12
3.0 Simulations	13
3.1 Simulation Definition	13
3.1.1 Stepped Approach	13
3.1.2 Research	14
3.1.3 Geometry	15
3.1.3.1 Simulation Scope and 2D Airfoil	15
3.1.3.2: Simplified LOTS	15
3.1.3.3 Borealis without Airbrakes	16
3.1.3.4 Borealis with Airbrakes	17
3.2 Stability Analysis and Initial Drag Tabulation	18
3.3 Final Verification	20
4.0 Conclusions	20
5.0 Addendum - Mesh Refinement Study	20
5.1 Purpose	20
5.2 Constraints and Criteria	21
5.3 Method	22
5.4 Results	24
6.0 Addendum 2 - Worst case stability study	25

Table of Definitions

Term	Hereafter referred as
Angle of Attack	AoA
Computer Aided Design (Model)	CAD
Coefficient of Drag	Cd
Computational Fluid Dynamics	CFD
Centre of Pressure	CoP
Leviathan of the Sky	LOTS

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Next, I wish to extend my thanks to Manav Toor, who has developed the simulation tool in Ansys and saved over 50 hours of labour in data processing and entering.

Lastly, I extend my heartfelt appreciation to the entire Waterloo Rocketry Team, for their unwavering support in the project.

0.0 Introduction

0.1 Problem Description

Recognizing that implementing airbrakes on a liquid fuel rocket has to reside within the vehicle's aerodynamic profile, specifically within the airbrakes section, which is between the Upper Body Tube and the Injector Bulkhead Tube, also known as the lower section of the upper body tube, presents a unique set of challenges. The placement of airbrakes in these areas is strongly suspected to reduce the rocket's stability, as it brings the center of pressure more towards the nose of the rocket. Effective implementation of airbrakes will require that their effect on the net rocket drag force be quantified. This report aims to analyze the impact of airbrakes on the team's 2024 rocket - Borealis using Computational Fluid Dynamics (CFD) simulations.

The interaction between the deployed airbrakes and the flow dynamics surrounding the rocket surface is complex. When extended, airbrakes alter the airflow patterns around the body of the rocket, potentially impacting the thermal and pressure gradients experienced by the rocket's structure. This alteration in flow could lead to unexpected aerodynamic behaviours, which are critical to understand to ensure the successful operation of the rocket.

The objective of this CFD analysis is to quantitatively assess these stability impacts. By simulating various deployment scenarios and configurations of airbrakes, including velocity, angle of deployment, physical shape and structure, and altitude (air density and viscosity), we aim to identify potential risks and develop strategies to mitigate them. This analysis will provide valuable insights into the aerodynamic behaviour of the rocket with airbrakes, helping to optimize their design and integration for enhanced safety and performance.

0.2 Background

In order to implement airbrakes effectively on the 2024 rocket, it is necessary to characterise their drag. The airbrakes have multiple degrees of freedom, since not only can they extend (thus changing the reference area and potentially the underlying drag behaviour), but they also operate over a wide range of airspeeds and altitudes. Thus, the drag force is a function of the drag coefficient (which is approximately a function of Mach number (which is a function of altitude) at high air speeds, as well as geometry) and reference area. Thus, we want to characterise airbrake drag along three axes - extension amount, altitude and airspeed. The governing equation for drag is:

$$F_D = \frac{1}{2} \rho v^2 C_D A$$

Where ρ , density of fluid, is dependent on the altitude, v , the velocity and A , the cross sectional area directly pulled out of SolidWorks.

The rocket is designed on the critical assumption that it will be pointing mostly straight up on ascent. The fins achieve this, applying corrective moments to the rocket to give it “static stability” during flight. An equivalent way to state this is that the fins shift the rocket's centre of pressure downward, below the centre of mass. See the 2023 flight dynamics report (

[Mini Report on Simulation Results \(Cycle 1\)](#)) for a more detailed explanation. Since the airbrakes are designed to impart drag forces and are placed well above the rocket's centre of mass, they risk shifting the center of pressure upward, potentially leading to instability if it is too close to or above the centre of mass. Validating the location of the centre of pressure is thus the other primary purpose of airbrake CFD simulations.

1.0 Scope and Requirements

1.1 Scope

The scope of the simulation is to take in rocket CAD (a 3D model of the rocket's outer surface) with airbrakes at various degrees of extension and return a (minimum) static stability margin and a net drag force.

The simulation must also closely examine the wake effect of the airbrakes and ensure it does not affect the direction and consistency of airflow over the fins very much with a close-to-final design of simple shapes. There is no easy quantitative measure of wake effects on fin effectiveness without a complicated fusion of CFD and dynamics simulations, so qualitative analysis of flow and pressure plots in the region around the fins is the primary method to validate wake effects. A testable requirement has not been set out for fin stability.

1.2 Constraints

As outlined by [Controls 2024 Timeline](#) : Requirements, WR-06, LC-R7.2.1, The rocket center of pressure, with the airbrakes fully extended, shall not extend less than 1.5 body calibres (9 inches) behind the center of mass, which will be determined by the simulation, and is hereafter referred to as the stability margin.

Simulation with surface roughness as a parameter can not utilize the GPU solver, hence it will further constrain our processing power.

1.2.2 Team Constraints

REQUIREMENT
Static Stability Margin: The primary objective is to maintain a baseline minimum stability of 1.5 body calibres in 95th percentile winds . This means that the centre of pressure (CoP) must always be at least 9 inches behind the centre of mass. CFD simulations will focus on ensuring that the deployment of airbrakes does not shift the CoP too far forward, which could lead to rapid destabilization of the rocket.

Deployment Phase of Airbrakes: Airbrakes will only be operational during the coast phase of the flight, post-engine burnout. This constraint is vital to avoid interference with the boosting phase of the rocket. Simulations will test the airbrakes' effectiveness and impact on the rocket's aerodynamics specifically during this coast phase, considering the maximum predicted velocity at burnout based on engine and mass data.

Velocity Range Coverage: The operational envelope for the airbrakes spans from the velocity at engine burnout to zero airspeed at apogee. Simulations must effectively cover this entire range, accounting for varying airspeeds and their impact on airbrake performance and rocket stability.

Simulation Verification: While Ansys simulations will be utilised for detailed analysis, it is important to approach the results with a critical eye. Over-reliance on Ansys for precise drag force calculations can be misleading. Therefore, drag force will also be approximated manually to cross-verify the simulation results. The aim is to stay within an **order of magnitude** of the manually approximated results. Additionally, the integration method to find the CoP can be unpredictable; thus, it will be validated through a combination of Ansys results and OpenRocket simulations.

1.2.3 Project Constraints

The robustness of the simulations are constrained by the 32 HPC licences sponsored by Ansys. Prof. Mike Cooper-Stachowsky has a lab of high-performance computers with limited availability, which will be used to conduct meshing and simulation with large compute requirements. Since only one person can use the computer at any one time, this will bottleneck our operations somewhat.

1.3 Interfaces

Airbrakes Control Software: The CFD simulations must provide the airbrake control system with Cd values over a range of velocities, altitudes, AoA, at different airbrake extensions.

Flight Dynamics: The CFD simulation will feed OpenRocket the aerodynamic drag on the rocket at various velocities, altitude, and angle of attack, in order to effectively override the OR calculation of drag.

Flight Dynamics: Simulations will take input data from each simulation cycle of Borealis (mass, geometry) and verify the worst case static stability margin

Flight Dynamics: The CFD simulation shall examine qualitatively the turbulent flow generated by the airbrakes over fins and other sections of the rocket, to ensure that such airflow will not have any considerable effect on the stability and nominal flight of the vehicle.

1.4 Timeline

📅 Controls 2024 Timeline

2.0 Apparatus

The CFD simulations are conducted in **Ansys Fluent**, which is a general-purpose computational fluid dynamics (CFD) software used primarily to model fluid flow.

SpaceClaim is an embedded 3D CAD design software by Ansys, it will be used to label named selections and create fluid enclosures, as well as the bodies of influence.

SolidWorks will be used to modify the geometry of the rocket being solved in Ansys.

OpenRocket will give us the reference value to compare results in Ansys to.

EDX | A Hands-on Introduction to Engineering Simulations is the comprehensive guide that is being referenced in many parts of the simulation.

2.1 Preparing Geometry

Simulation starts by importing the geometry of the rocket, with the interior completely solid and watertight, into Spaceclaim. This model is then checked for the location of the origin (it should be along the main axis of the rocket, so the center of pressure is integrating over the correct region).

The rocket is also rotated so one of the planes defined by the x,y,z global coordinate slices directly through one of the fins and the halfway point between the other two fins (and center of the airbrake). This way when a plane is created for drawing streamlines, it will be correctly oriented.

The rocket will have BOIs created at the nosecone, airbrakes, fins, and boattail. Such areas are where turbulent flows are generated and have the most “awkward” geometry on the rocket, therefore it's crucial to mesh such regions to a higher resolution.

The list of BOIs will be:

- nosecone close
- nosecone far
- airbrake panels
- fin close
- fin far
- rocket body

Rocket body will be a cylindrical BOI just slightly bigger than the diameter of the rocket, and encapsulates the entire rocket except the fins.

Nosecone close and nosecone far will be BOIs taking the shape of a cone and encapsulates the frontal area (nose cone) of the rocket.

Fin close will contain the transition area from the fin to the rocket body. Fin far will capture the rest of the fins.

Airbrake panels will capture all three panels of the rocket.

When creating the BOIs, it is crucial to ensure that no merge is turned on, so the rocket geometry remains and does not get merged with the BOI.

Once all the BOIs are created, a cylindrical enclosure is created around the rocket, with 150 inches in front, 850 inches behind and 350 inches radially from the rocket. This distance was later validated in post processing that the boundary walls are far enough to not affect airflow over the rocket. The front of the enclosure is named “inlet”, back of the enclosure “outlet”, outer wall “walls” and interior of the rocket “rocket”, respectively. Again the rocket has physics suppressed, therefore when an enclosure is created in the rocket, it essentially models the airspace around the rocket, with the rocket carved out as “void”, or vacuum.

2.2 Meshing

The above geometry is then carried over to the Fluent mesher, with units expressed in inches. Values from the mesh refinement study are filled in, the geometry is described to contain both fluid and void, with the rocket body itself considered to be void. Inlet, outlet and walls are set accordingly, and topology is left as default unchanged.

The surface mesh is created first, with the maximum skewness hovering usually between ~0.6-0.7, which is perfectly acceptable. Then the volume mesh is created with polyhedral units. Once the mesh is fully resolved, the “verify mesh” function is used to check the integrity of the mesh.

If no error arises, the skewness, orthogonality and aspect ratios are checked, if they are within our tolerance, the mesh is brought over to the Ansys solver for solving (later steps are often invoked through PyAnsys, but for the purpose of documentation, a detailed procedure for how it is done in the Ansys Fluent GUI will be recorded).

2.3 Solving

Mesh transfer to the Ansys Fluent solver is fully automatic. Fluent will check the mesh and open it up if no errors occur. At this time it is important to double check that only single precision is

used for the solver, and (n-1, where n is the total number of threads available to the CPU) threads are used, up to the limit of HPC cores available to Ansys (at the time of writing, 32).

Starting by setting the air property. For the altitude being solved for, get the dynamic viscosity and density from the Atmospheric Properties Calculator located at <https://aerospaceweb.org/design/scripts/atmosphere/>. Boundary equations are set with the inlet as the velocity of the rocket, with the direction of inlet parallel to the rocket and in the opposite direction (pointing towards the tail). Angle of attack is added by using trig and resolving the velocity with angle into components. Outlet is set to be a pressure outlet with a gauge pressure of 0.

To resolve the velocity components with angle of attack, given an angle of attack α , formula is:

$$Velocity_{parallel} = Velocity_{total} * \cos(\alpha)$$

$$Velocity_{perpendicular} = Velocity_{total} * \sin(\alpha)$$

The enclosure wall is set to be free slip, which means the shear stress is 0pa everywhere, allowing airflow to be undisturbed by the wall. The rocket wall is set to a specific roughness according to the final surface finish of the rocket, roughness constant is between 0.5-1, with 0.5 being perfectly uniform throughout. This value will be matched with OpenRocket surface roughness, 0.5 roughness constant is used as this is the way it is predicted in OpenRocket.

The system model used is shear stress transfer (SST) k-omega. It combines the advantages of the $k-\epsilon$ and $k-\omega$ models to improve accuracy in predicting the onset and amount of flow separation. The governing equations are:

Kinematic Eddy Viscosity

$$\nu_T = \frac{a_1 k}{\max(a_1 \omega, SF_2)}$$

Turbulence Kinetic Energy

$$\frac{\partial k}{\partial t} + U_j \frac{\partial k}{\partial x_j} = P_k - \beta^* k \omega + \frac{\partial}{\partial x_j} \left[(\nu + \sigma_k \nu_T) \frac{\partial k}{\partial x_j} \right]$$

Specific Dissipation Rate

$$\frac{\partial \omega}{\partial t} + U_j \frac{\partial \omega}{\partial x_j} = \alpha S^2 - \beta \omega^2 + \frac{\partial}{\partial x_j} \left[(\nu + \sigma_\omega \nu_T) \frac{\partial \omega}{\partial x_j} \right] + 2(1 - F_1) \sigma_{\omega 2} \frac{1}{\omega} \frac{\partial k}{\partial x_i} \frac{\partial \omega}{\partial x_i}$$

Closure Coefficients and Auxiliary Relations

$$F_2 = \tanh \left[\left[\max \left(\frac{2\sqrt{k}}{\beta^* \omega y}, \frac{500\nu}{y^2 \omega} \right) \right]^2 \right]$$

$$P_k = \min \left(\tau_{ij} \frac{\partial U_i}{\partial x_j}, 10\beta^* k \omega \right)$$

$$F_1 = \tanh \left\{ \left\{ \min \left[\max \left(\frac{\sqrt{k}}{\beta^* \omega y}, \frac{500\nu}{y^2 \omega} \right), \frac{4\sigma_{\omega 2} k}{C D_{k\omega} y^2} \right] \right\}^4 \right\}$$

$$C D_{k\omega} = \max \left(2\rho\sigma_{\omega 2} \frac{1}{\omega} \frac{\partial k}{\partial x_i} \frac{\partial \omega}{\partial x_i}, 10^{-10} \right)$$

$$\phi = \phi_1 F_1 + \phi_2 (1 - F_1)$$

$$\alpha_1 = \frac{5}{9}, \alpha_2 = 0.44$$

$$\beta_1 = \frac{3}{40}, \beta_2 = 0.0828$$

$$\beta^* = \frac{9}{100}$$

$$\sigma_{k1} = 0.85, \sigma_{k2} = 1$$

Where F1 blends the two models near the wall and F2 blends the two models in the free stream.

The SST k- ω turbulence model is particularly suitable for modeling airflow over the exterior of a sounding rocket due to its accurate prediction of flow separation under adverse pressure gradients, enhanced near-wall treatment, and robustness across a wide range of flow conditions. Its hybrid approach combines the strengths of the k- ω model near the wall and the k- ϵ model in the free stream, ensuring precise calculations of drag and center of pressure. This model's capability to handle varying aerodynamic environments, from subsonic to supersonic flows, makes it ideal for comprehensive analysis of the rocket's aerodynamic performance.

The model solves two transport equations: one for the turbulent kinetic energy (k) and one for the specific dissipation rate (ω). These equations describe how turbulence is generated, dissipated, and transported throughout the flow field.

A quick check on the reference frame, specifically the global origin, is done to ensure a known location to base off center of pressure calculations.

Reference values will be computed from inlet, with the reference zone being the air surrounding the exterior of the rocket, length is set to the length of the rocket, and area set to the frontal cross section area of the rocket, both values are computed from OpenRocket and Solidworks.

Solution method will be coupled for pressure-velocity coupling, with least squares cell based gradient and second order upwind for every other discretization.

Under report definitions, two reports are created. First being the drag force report, which is set with a vector parallel and in the opposite direction of the rocket and on the rocket body. Second is the center of pressure report, which is computed as:

$$CP_x = \text{arealnt}(X * \text{Pressure})@\text{rocket} / \text{arealnt}(\text{Pressure})@\text{rocket}$$

$$CP_y = \text{arealnt}(Y * \text{Pressure})@\text{rocket} / \text{arealnt}(\text{Pressure})@\text{rocket}$$

$$CP_z = \text{arealnt}(Z * \text{Pressure})@\text{rocket} / \text{arealnt}(\text{Pressure})@\text{rocket}$$

The location of concern is the axis parallel to the rocket, the other two values should evaluate very closely to 0.

Hybrid initialization and residuals are set accordingly depending on the precision requirement, and the calculation is run until convergence.

2.4 Solution and Post Processing

The solution is opened up in Ansys Post, planes are drawn on different axis and cut through the rocket at different angles. Velocity streamlines are drawn both forward and back, flows over airbrakes and fins are evaluated to determine if there are any concerning patterns.

3.0 Simulations

3.1 Simulation Definition

3.1.1 Stepped Approach

With limited knowledge in fluid dynamics, it is difficult to tell whether Ansys is giving the ground truth, or just spitting out random numbers that seemingly make sense, but in reality are nothing close to the true answer.

This is further complicated since, with the coefficient of drag (C_d) mainly ranging from 0-1 (in most cases) and CoP being anywhere between 0->~4m (length of the rocket being modelled), it is easy to get false values that seem reasonable. Unless the number intuitively does not make sense, the simulations will need to be validated against a second source, before being deployed to production. In other words, we need to have a sanity check.

Ansys has numerous parameters which are used in calculation, but are not explicitly called out when running through the simulation procedure - if they are not changed, Ansys will use default values.

As a preliminary sanity check on our basic understanding of Ansys Fluent, simple geometries were modelled in Ansys and solved first. They are simple enough shapes to allow manual approximation of drag under given conditions to a reasonable degree of precision. With such geometry modelled and solved, the answer was compared against the hand-calculated values. Drag coefficients of simple bodies are given below in Figure 1.

Type of body		Reference area S	Reynolds number Re	Drag coefficient C_D
Cube		$S = D^2$	$Re > 10^4$	1.05
		$S = D^2$	$Re > 10^4$	0.8
Solid hemisphere		$S = \pi D^2 / 4$	$Re > 10^4$	→ 0.42 ← 1.17
Hollow hemisphere		$S = \pi D^2 / 4$	$Re > 10^4$	→ 0.38 ← 1.42
Thin disk		$S = \pi D^2 / 4$	$Re > 10^3$	1.1
Circular disk		$S = \pi D^2 / 4$	$Re \leq 1$ $Re > 10^4$	$20.4/Re$ 1
Sphere		$S = \pi D^2 / 4$	$Re \leq 1$ $1 < Re \leq 2 \times 10^5$ $Re > 2 \times 10^5$	$24.0/Re$ 0.45 0.2
Streamlined body		$S = \pi D^2 / 4$	$Re > 10^5$	0.04

Figure 1: drag and other properties of simple geometries

With the solver set up and confirmed to output correct results for a simple geometry, the solid model of LOTS (the team's 2023 liquid rocket) was analysed, with the same setup as the sanity check (meaning the meshing method, solver model, boundary conditions) with minimum

changes. The model was the surface of the LOTS rocket CAD with 100% solid infill. The results of this simulation were compared against the following alternative sources:

- Cycle 6 LOTS OpenRocket Simulation
- Hand calculations

The LOTS model was simplified to encompass only the essential geometry, including a singular extrusion from the nose cone to the fin can, and incorporating the boattail as a single solid entity. Minor features like fasteners were omitted to simplify the analysis.

The Ansys simulation results for LOTS were deemed verified if they exhibit a strong correlation with OpenRocket outcomes, based on the following criteria:

- **Drag within 5% deviation from OR**
- **Centre of Pressure within 5 cm from OR**, favouring a rearward positioning to account for base drag not considered in OpenRocket.

Upon successful verification of the LOTS model, the focus shifted to developing a more intricate geometry that aligns closely with the team's 2024 liquid rocket, "Borealis". Given the preliminary stage of Borealis' design, the following assumptions are incorporated into this analysis:

- There will be 3 fins clocked at 0, 120, 240 degrees
- The total rocket length will be pulled from the most updated flight dynamics simulation cycle
- The geometry of the fins will be identical to LOTS
- The airbrakes will have 3 panels, clocked at 60, 180 and 300 degrees so as to offset them from the fins

This model will produce the final guiding results for control/flight dynamics usage.

Recognizing that flight dynamics predictions vary between cycles, and it is impractical to rerun sims for the entire rocket body each time a flight parameter is changed, the airbrakes simulation shall run for **stability (CoP) analysis only** when:

- A flight parameter gets modified, such as change in rocket length, weight distribution, new geometry, airbrake panels resize, etc.

The airbrakes simulation shall be run in **full scope** for:

- Initial model verification and producing results for other systems to work from
- Before the final flight to produce the dataset that will go live onboard.

Where full scope is defined as simulation that fully characterises a rocket's stability and drag under a range of conditions.

3.1.2 Research

The easiest way to collect data was to research past CFD simulations in other design teams' reports. However, it is difficult to validate the reliability of such info as the detailed methodology and simulation setup they used may not be well documented in such a report, unless multiple team reports back up on the same approach. Some sources are presented here:

“Stability Analysis of Conventional Rocket Model using CFD Tool”

<https://www.ijert.org/research/stability-analysis-of-conventional-rocket-model-using-cfd-tool-IJERTV9IS010262.pdf>

Ansyz forum post about CoP calculation

<https://Ansyskm.Ansys.com/forums/topic/how-is-the-center-of-pressure-calculated-in-Fluent/>,
<https://www.afs.enea.it/project/neptunius/docs/Fluent/html/ug/node962.htm>

3.1.3 Geometry

3.1.3.1 Simulation Scope and 2D Airfoil

The testing sim will contain a 2D airfoil and a simplified model of LOTS. Figure 2 shows the geometry of the airfoil in DesignModeler.

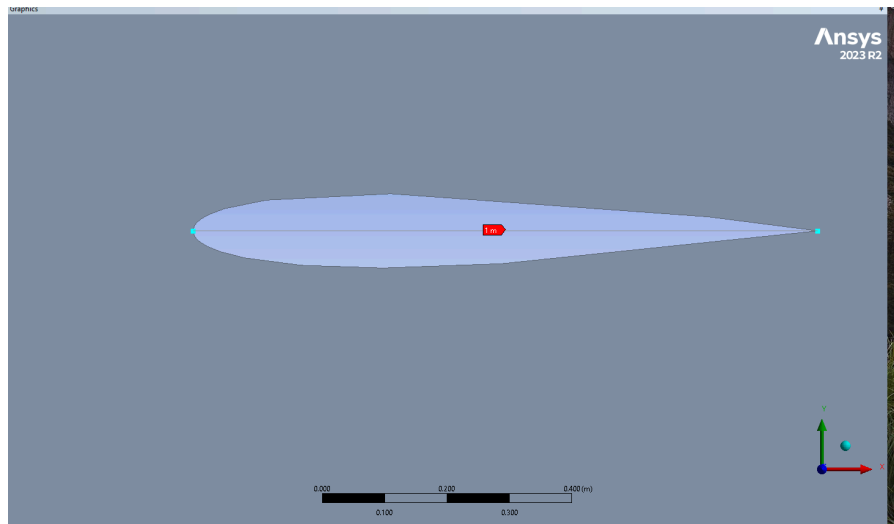


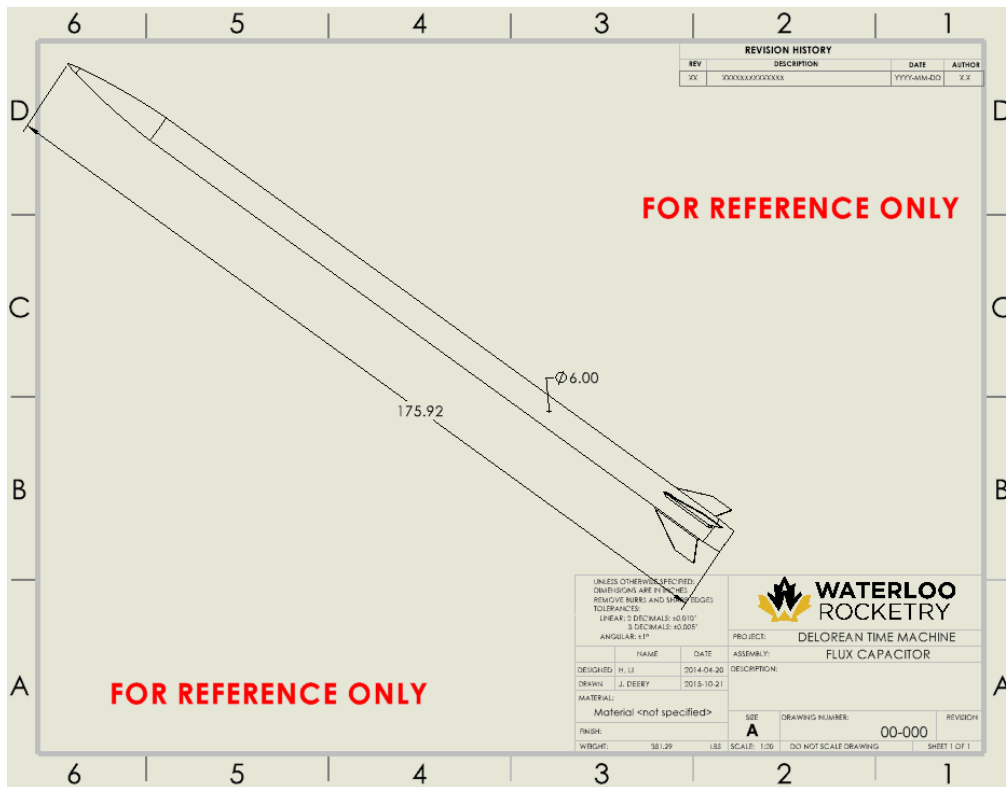
Figure 2: Geometry of an airfoil in DesignModeler

3.1.3.2: Simplified LOTS

The geometry of the “simplified LOTS model” is shown in figure 3.



Figure 3: Simplified



This model reduces the geometry of the LOTS rocket to a single extrusion between the nose cone and the boattail of the rocket. The geometry of the nosecone and the fins are preserved, through directly modifying the existing LOTS CAD.

Small details, such as bolt circles, CAN passthrough over the Oxidizer tank, and irregularity of the nose cone tip are not preserved in this CAD.

3.1.3.3 Borealis without Airbrakes

Figure x is an indicative CAD of the potential length/geometry of Borealis. Although it is still adapting the nose cone equation and fin shape of LOTS, the length of the rocket is updated, and minor details including a spherical area over the tip of the nose cone, and top face geometry of the fins, are added.

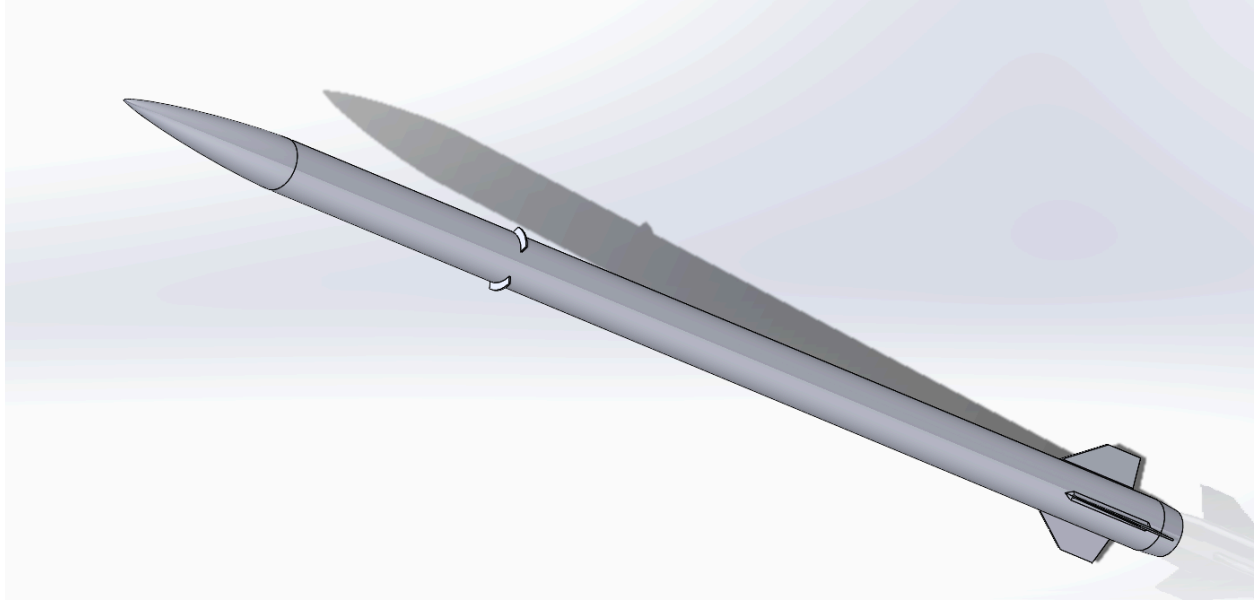


3.1.3.4 Borealis with Airbrakes

The airbrakes are three hemispherical plates, perpendicular to the rocket's body and with the horizontal centerline of each plate offset from the neighbouring two fins at 60 degrees.

All three of the airbrakes are planer, identical in size and shape, and are constrained to extend the exact same amount for x rotation of the motor.

The airbrakes are located at 58.65 in from the tip of the nose cone, with a length of 0.86 in when fully extended, and take the shape of a rectangle with a semicircular cutout in one end. The radius of curvature for this semicircle is 3 inches, so when it fully closes it perfectly aligns with the profile on the rocket's body tube.



3.2 Automation with Python

In order to reduce the total number of man hours configuring and running each individual simulation, it was decided to use the Ansys Python API, PyAnsys, to automate all the simulations. This allowed for minimal human intervention and error as well as allowing the team to run multiple simulations in batches overnight using a university provided supercomputer.

Using PyAnsys in no way modified or altered the simulation data, this simply automated and sped up the process, allowing for the team to collect a larger amount of data points and construct more accurate plots.

The program allows a user to input all meshes into a folder (each mesh represents a different extension of air brakes) and update a configuration file for all necessary parameters. The current configurations used for the simulation are altitude, velocity, and angle of attack. Using this information, the program also calculates air density and viscosity using the 1976 U.S. Standard Atmosphere. All other values (rocket length, convergence values, etc.) are constant but can be modified by altering csv files in the configurations folder of the program.

Once all input cases have been added, the user can hit run and leave the computer running overnight. Every input case will be run on every mesh and all data will be outputted into a csv file for easy use. Furthermore a velocity and pressure contour of each simulation case is saved into a directory as well. This is helpful to visualize any discrepancies.

The program can be found at <https://github.com/waterloo-rocketry/PyAnsys/>

3.3 Airfoil

The modelling and verification of the airfoil follows exactly Module 5: Computational Fluid Dynamics (CFD) - 5.1 Turbulent Airfoil Flow of the EDX CFD course with close to identical results, hence our software is validated.

3.4 Stability Analysis and Initial Drag Tabulation

Using rocket geometry in 3.1.3.3 and 3.1.3.4, initial verification is done through cross-comparing with Openrocket. Rocket data are directly taken from the cycle two simulation of 2024 Borealis rocket: [Cycle 2 Summary of Results](#) . The details of the simulation are as follows.

Velocity of the rocket (m/s) vs Drag Force (N) at 0, 50, 100% extension

Velocity (m/s)	0%	50%	100%
34	5.347183	6.28797	7.931131
68	20.07957	24.0438	30.7105
102	43.62593	52.82345	67.56419
136	75.70178	92.39534	118.7119
170	116.1482	142.6004	183.432
204	164.7873	203.3864	264.2714
238	221.5784	274.6371	357.3065
272	286.4355	356.2869	464.7654
306	359.1301	448.3496	585.665
340	439.8027	550.5583	720.5277
374	528.4389	663.197	868.7817
408	624.8264	786.1453	1028.554
442	729.0035	919.0902	1205.442
476	840.8751	1062.47	1393.69
510	960.4208	1215.985	1595.505

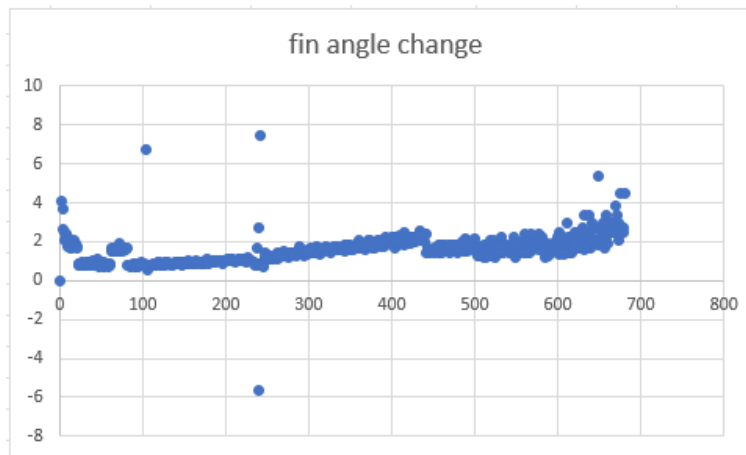
544 1087.722 1379.528 1814.117

The center of pressure of the rocket is also investigated for each of the above case, which resulted in the worst case stability study (Addendum 2), using the most recently updated cycle 2 geometry of Borealis with a length of 4.105m and cross-sectional area of 0.0232 square meters, the center of pressure is at 0.9853936 m ahead of the boattail, or 3.1196064 m behind the nose of the rocket, under the condition of 7000 m (22965 ft), 34 m/s upward velocity at 0 degrees of attack (Mach 0.1) and 100% airbrake extension.

This center of pressure value is within 2% deviation of an OpenRocket simulation of the same set of values.

By graphing the streamlines and pressure contours on section planes that cut through the rocket at different angles, and carefully observing the velocity streamline over the fins after hitting airbrakes, it has been visually verified that the turbulent flow over the airbrakes does not *significantly* impact the airflow over the fins to a level of concern.

To double check airflow over fins, fin angle change in the time it takes for the fins to reach where the airbrakes once were = roll rate * airbrake to fin vertical distance / vertical velocity. The fin angular distance never gets above 5deg, and that only happens at the end when fins matter less. so long as the airbrake wake angular width is not larger than 50deg (60 deg separation - 5 deg on both ends), airflow over the airbrakes does not affect the airflow over the fins.



3.5 Final Verification

For the final drag tabulation, the solver was refined to steps of 10% extension, which were simulated across velocity of Mach 0.1 to 1.6 in 0.1 increments and altitude of 0m to 10 000m in 1000m increments. For simplicity by avoiding another domain of simulation, the rocket is always assumed to travel upright orthogonal to the ground reference frame, or angle of attack of 0.

4.0 Conclusions

Center of pressure pulled from cycle two OpenRocket is 304 cm behind the nosecone, which is within 5% margin of Ansys results, therefore Ansys results are deemed reasonable and successfully validates that airbrake's effect on the stability of the rocket is within a reasonable level. With fin optimization, the stability remains above 2 calibers at all times.

Drag curves are reasonable and show a direct and clear relationship between drag, velocity, altitude and extension rate, which follows intuition and corresponds to OpenRocket values at multiple data points.

5.0 Addendum - Mesh Refinement Study

5.1 Purpose

To effectively reduce computation time and increase solving efficiency, i.e. solve more models in the same amount of time and produce more data points, it is necessary to conduct a mesh refine study, in order to determine the minimum amount of meshing cells required to solve a given model, while giving an output to a degree of accuracy that shall not have deviations that impacts the application of the results.

5.2 Constraints and Criteria

H0: "Decreasing the mesh refinement (increasing element size) does not significantly reduce the accuracy of the simulation results."

H1: "Decreasing the mesh refinement (increasing element size) significantly reduces the accuracy of the simulation results."

Accepting H0: This would mean the changes in mesh refinement (making the mesh coarser) do not significantly reduce the accuracy of the results. In practical terms, the study concludes that a coarser mesh is sufficient for your simulation needs, balancing computational efficiency with acceptable accuracy. H0 will be accepted if the refined/optimised mesh is <0.5% deviation from the finest mesh, as defined below.

Reject H0: This implies that decreasing the mesh refinement significantly reduces the accuracy of the results. In this case, a finer mesh would be necessary to maintain the desired level of accuracy in the simulations.

The finest mesh is constrained by the computational power of the AI computer ECE-NEBULA01 with the following specifications:

CPU	GPU	RAM
AMD Ryzen Threadripper PRO 5955WX 16-Cores 4.00 GHz	Nvidia RTX A6000	128 GB (128 GB usable)

The finest mesh is further constrained by the Ansys licensing limitation, which is 32 logical processors.

The finest mesh shall be computed within a reasonable timeframe, which is arbitrarily decided to be 30 minutes, since we are simulating a simple body.

The finest mesh shall converge or reach a residual of $< 1e-3$ with the average iteration of < 5 seconds. This ensures that if the model is computed to 400 iterations (in case it doesn't converge before such), it will at worst have a computing time of 33 minutes.

Realistically, the model shows signs of convergence at ~ 100 iterations.

There will be three models in total: Finest, fine and rough. The two finest models will first be computed. Linear interpolation gives the average result that shall be used as the baseline. Then the rough model is computed and adjusted until the results fall in $< 0.5\%$ deviation.

The study will not deal with symmetry. However when set up correctly, it will drastically speed up the solving process. When

All meshes must satisfy the mesh quality chart shown in figure x.

Maximum skewness must strictly be under 0.95, with minimum orthogonal quality above 0.1. Both skewness and orthogonal quality must be in the very good section of the spectrum in worst cases, with Excellent being the preferred.

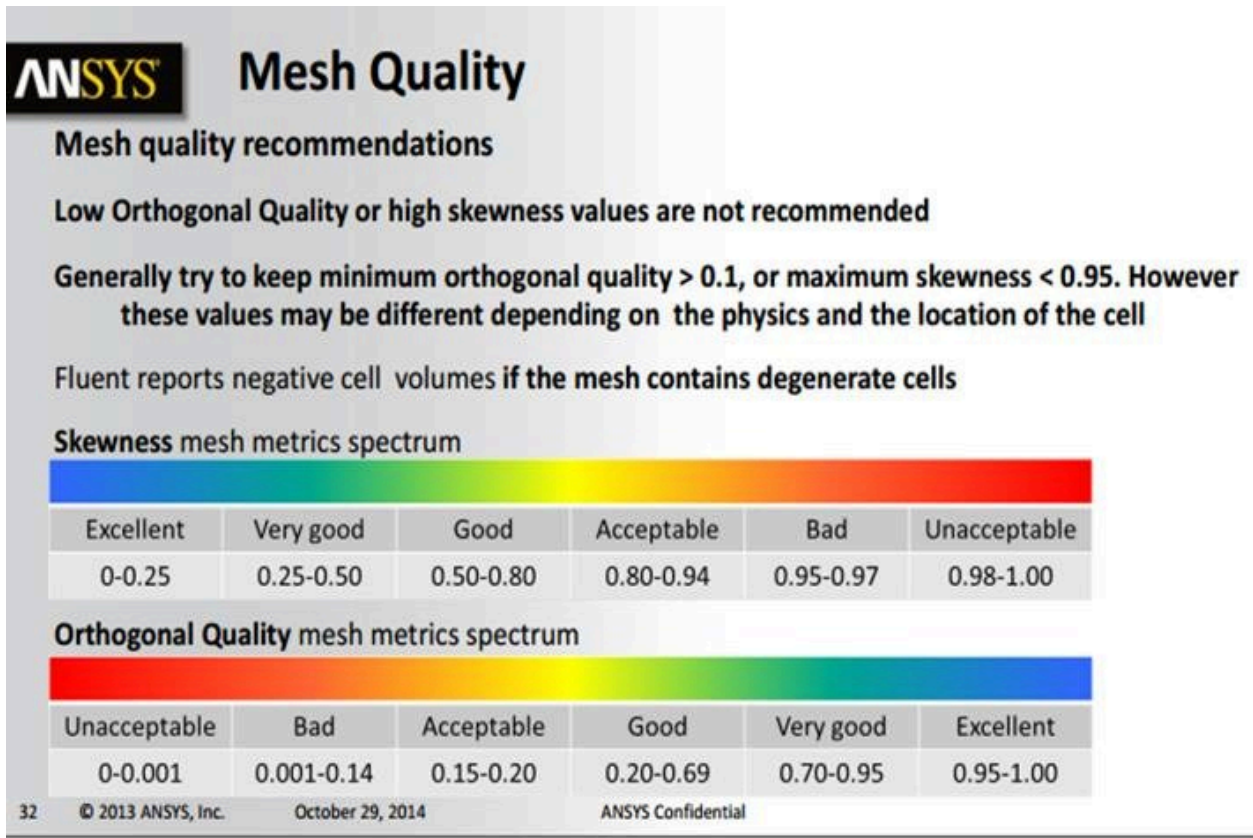


Figure x: Ansys suggested Mesh quality

5.3 Method

The method will be detailed in the main report.

All simulation will be conducted with the LOTS 2023 model, with the reference values shown in figure x.

Reference Values

Area [m ²]	0.018241
Density [kg/m ³]	1.225
Enthalpy [J/kg]	0
Length [m]	0.152
Pressure [Pa]	0
Temperature [K]	288.16
Velocity [m/s]	274.4
Viscosity [kg/(m s)]	1.7894e-05
Ratio of Specific Heats	1.4
Yplus for Heat Tran. Coef.	300

Reference Zone
enclosure-enclosure

Figure x: Reference values used for mesh refinement study

Other values, such as wall roughness, are left constant in all simulations, thus it will be abridged from the report.

Enclosure size, Body of Influence (BOI) shape and size will be kept consistent throughout the study.

Cases are run with the following data:

[in]	Fin BOI	Nosecone BOI	Surface mesh size	Transition layers
Case 1	0.8	0.6	0.1-45	10
Case 2	0.6	0.5	0.1-30	10
Case 3	0.5	0.4	0.1-20	10
Case 4	0.05	0.05	0.05-46.8	10
Case 6	0.02	0.035		10
	0.015	0.025	0.015-46.8	10
Case7	0.02 (base)	0.02 (close)	0.015-46.8	10
	0.015 (far)	0.2 (far)		

Each model will run 200 iterations, which should sufficiently converge the answers. The residuals (continuity, x-velocity, y-velocity, z-velocity k, omega), and drag force will be compared and averaged.

5.4 Results

The computations started with tetrahedral mesh, however there were no direct increase in quality gain with exponential growth in mesh sizes and cell, as shown below.

Case	Volume mesh method	Max cell length	Cell count	Average Orthogonality	Average Skewness	Minimum Orthogonality	Maxmium Skewness
1	Tetrahedral	50	2600000	0.78978	0.20913	0.08634	0.91366
2	Tetrahedral	30	3011493	0.79038	0.2085	0.087	0.913
3	Tetrahedral	25	4295589	0.8069	0.19154	0.05026	0.94974
4	Tetrahedral	30	37250044	0.88203043	0.197	0.16	0.972
5	Tetrahedral	64	21608498	0.8154069	0.18327231	0.11008753	0.88991247
6	Tetrahedral	64	39944573	0.81305	0.18568	0.101	0.89845
7	Tetrahedral	60	34142356	0.891	0.184	0.176	0.928

However, even though the gain was marginal, the computation time was exponential. At case six, the time to compute for a single iteration is well above 20-30 seconds, which far exceeds the under 30 minutes to convergence required, and is very suboptimal for large scale calculations, therefore, significant changes were required to the mesh.

Following advice of a team member (Cristian Bicheru), the mesh method was turned to polyhedral with double precision turned off. Tetrahedral meshes tend to create more numerical diffusion, especially around complex geometries like airbrakes, due to their less effective representation of curved surfaces and flow features. Polyhedral meshes, on the other hand, offer better numerical stability and solution accuracy in these regions due to their higher face counts and improved ability to conform to complex shapes.

By switching to polyhedral, we could potentially reduce the computational time as these meshes handle the fluid dynamics around the airbrakes more efficiently. Additionally, turning off double precision, while it slightly reduces the numerical precision of the simulation, can significantly decrease memory usage and computational overhead, speeding up the overall process.

A study was also done on curvature normal angle and cell per gap, which did not find any direct correlation between it and the quality of the mesh.

Curvature normal angle	Cells per gap		avg skewness	max skewness
18	1		0.03345	0.856
25	1		0.034317493	0.85615376
40	1		0.035648003	0.85615376
18	3		0.032825841	0.85615435
18	5		0.033631962	0.85615566
25	5		0.033664353	0.85615566

Therefore, the following values were used to conduct the polyhedral mesh:

CaseA	Fin BOI	Fin Close BOI	Nosecone BOI	Nosecone Close BOI	Airbrake BOI		Surface mesh size
	0.03	0.054	0.15	0.04	0.1		0.03-51.89
	0.07	0.04	0.15	0.05	0.1		0.03-46.63

The cell number dropped from 34 mil to 1.3 mil with average orthogonality of 0.16 average skewness of 0.9. the skewness sitting around 0.01-0.2, max skewness of 0.70. Upon inspection

of individual cells, all the highly skewed cells are in air cells distant away from the rocket and none actually particularly concerning the rocket. Areas of concern, including fins, nosecone, airbrakes all have a mesh in the Excellent region of Ansys' suggestion. Therefore, the mesh refinement study concludes with the above set of values used for final solve.

6.0 Worst case stability study

https://docs.google.com/spreadsheets/d/1yrRCH4vebvrrPBrgjtCP7KK3ymbRddPD5F0F_Y44o-U/edit#gid=2008932162

For understanding the worst case, the program is run for 150 iterations for each case. For the worst case, the program is run for 1000 iterations which achieves a convergence of at least $1e-7$ for all computed values except $1e-4$ for continuity.